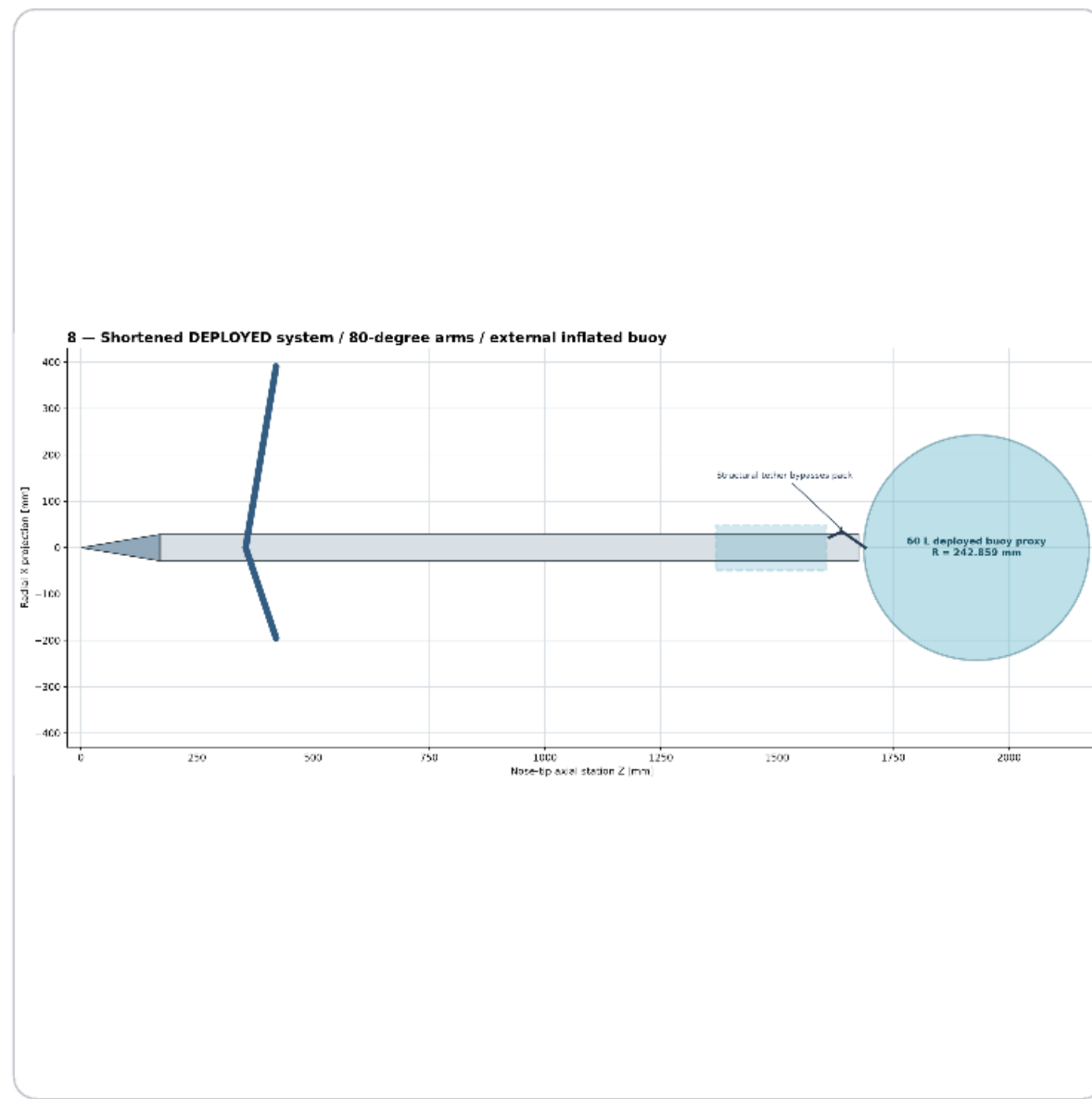


STINGRAY digital product definition

The mechanism is digitally coherent. The inflation architecture is not releaseable.

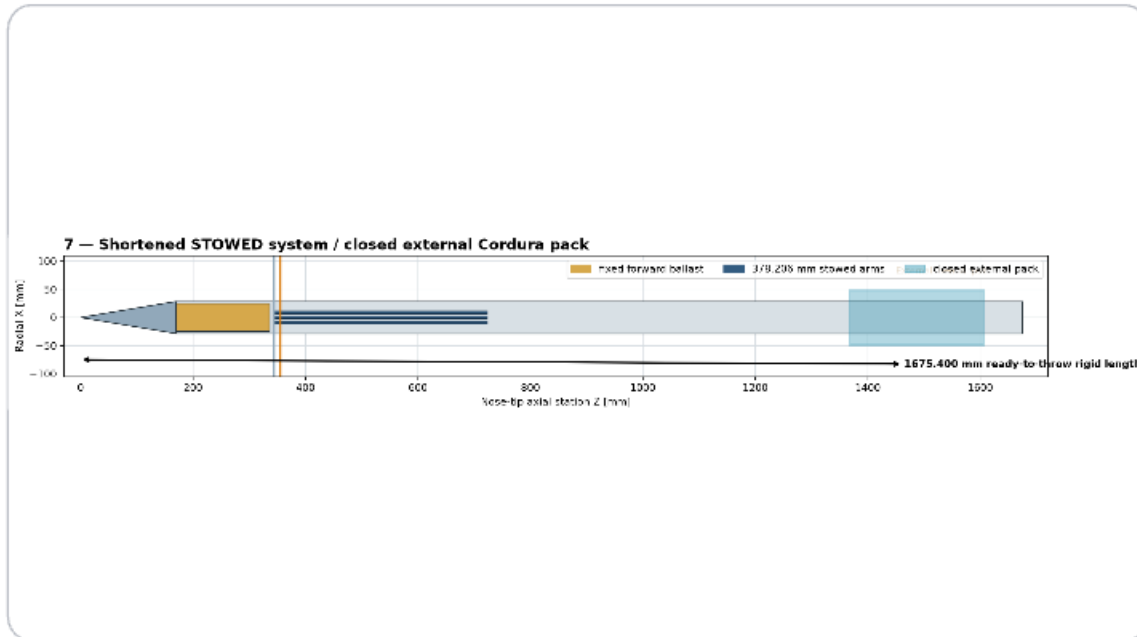
NOT RELEASED FOR FABRICATION OR FIELD USE

Frozen configuration: 1,675.4 mm rigid body | 355 mm arm pivot | external 60 L buoy pack

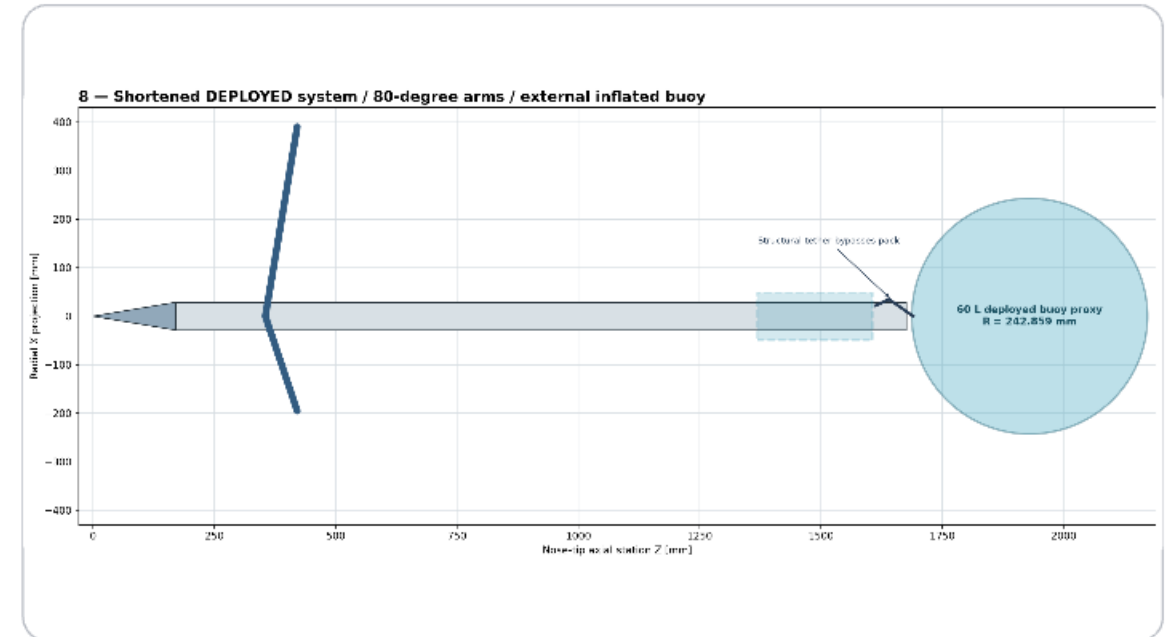


A compact body deploys arms and an external recovery buoy

The system is intended to remain compact before use, deploy three arms, open an aft softgoods pack, and transfer recovery load through a dedicated tether path.

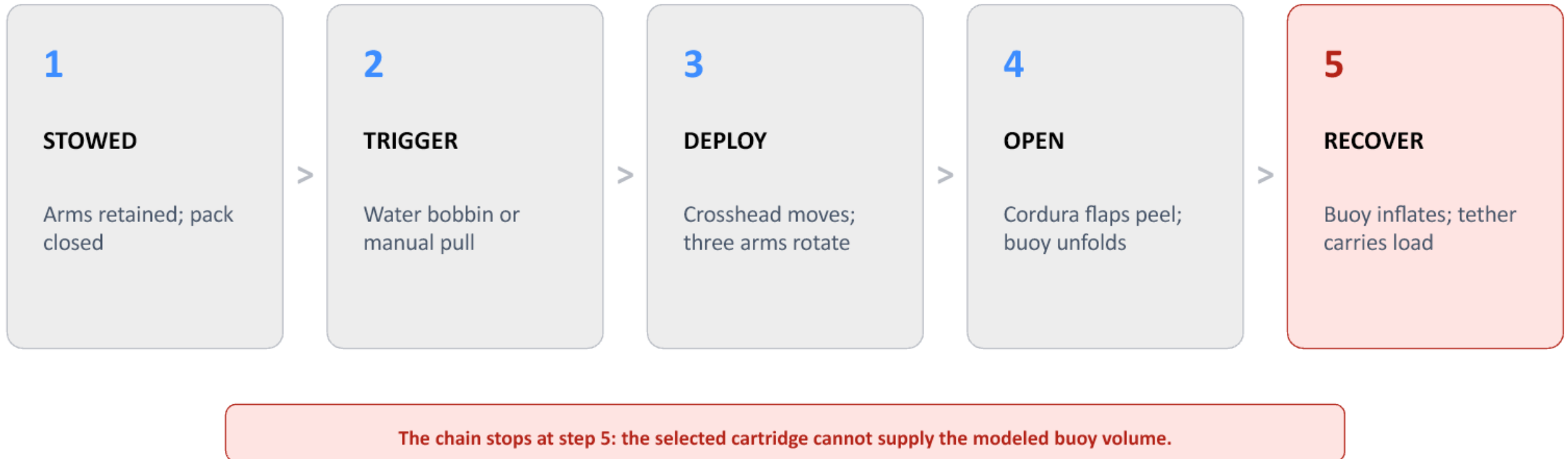


STOWED - 1,675.4 mm rigid length



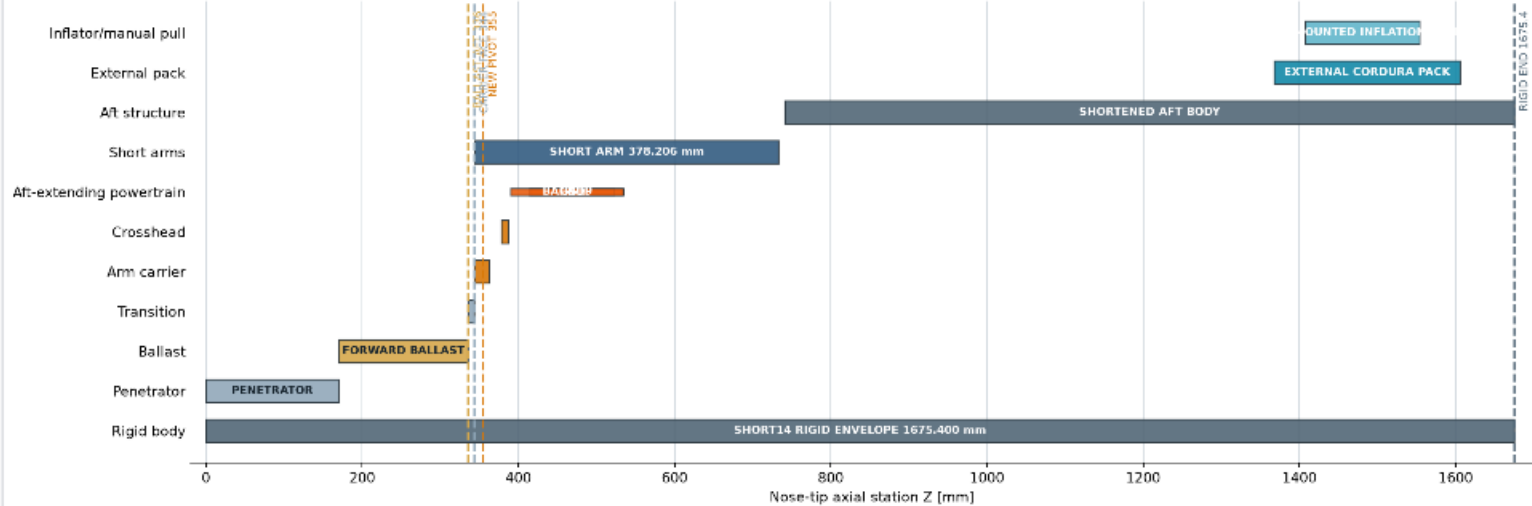
DEPLOYED - 80 degree arms, open pack

Operation is a five-step chain - and every link must work



Three subsystems share one continuous mechanical load path

2 — Corrected axial side section: true forward carrier and external buoy pack



FORWARD BODY

Penetrator and ballast establish the nose-side structural datum.

ARM POWERTRAIN

Gas springs, dampers, common crosshead, links, and stops rotate three arms.

AFT RECOVERY

Closed/open softgoods, inflator, buoy harness, hardpoint, and HMPE tether.

The frozen digital assembly is complete enough to audit rigorously

Every claim below is tied to the frozen AP242 masters and their authoring inventories.

180

named occurrences in each
endpoint state

92

unique part definitions

81

modeled arm positions from
0 to 80 degrees

697,167

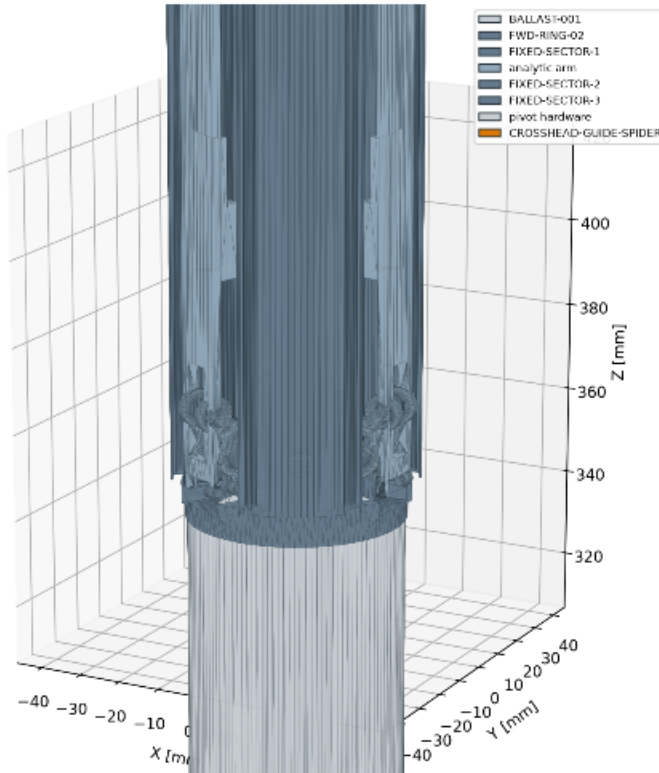
full-motion exact pair rows

ZERO unauthorized rigid overlaps | ZERO blocked Booleans | ZERO track/fit-register errors

Mass: 10.583 kg | reserve to 18.14 kg limit: 7.557 kg | rigid length: 1,675.4 mm

Digital mechanism motion passed every modeled position

Exact forward-ballast aft face Z=336 / 8 mm transition / carrier face Z=344 / pivot Z=355



Technical evidence view generated from frozen scripted B-rep source; not Creo evidence. Units: mm.

0-80°

one-degree audit coverage

15.055

mm crosshead travel

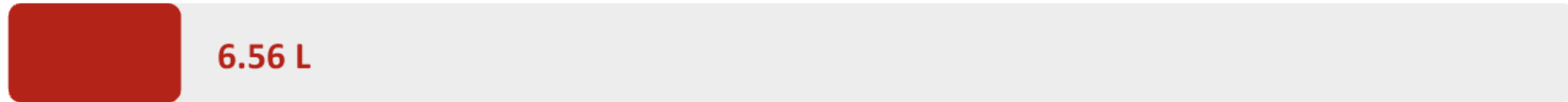
- 43,035 rows in the five-angle gate
- 697,167 rows in the full sweep
- closure residual: 2.49e-14 mm
- no invalid or unclassified rigid interference

Digital PASS does not prove deployment time, loads, fatigue, or physical lock capacity.

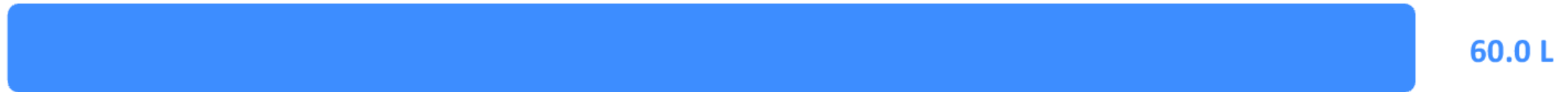
The selected 12 g cartridge can supply only about 11% of the modeled buoy

An optimistic ideal-gas calculation already proves the mismatch. Real delivery would be lower because of cooling, residual gas, pressure drop, leakage, and required buoy pressure.

Selected cartridge at 20 C



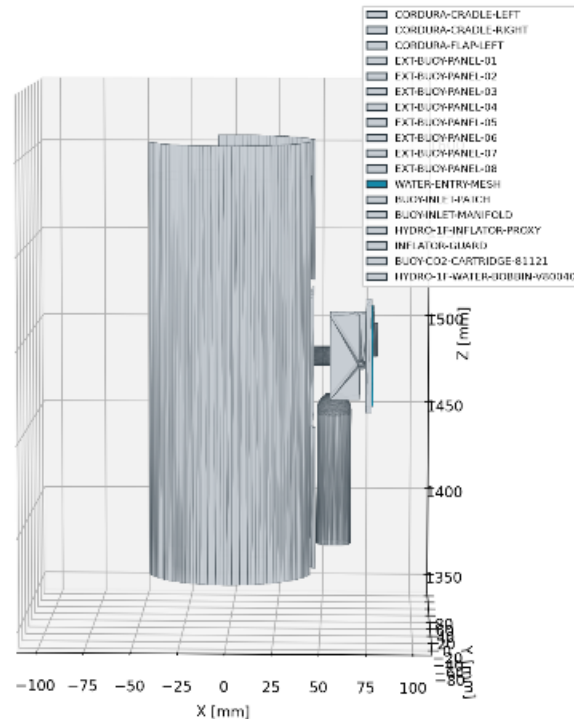
Modeled buoy nominal volume



FAIL: ideal lower-bound CO₂ mass is 109.8 g, before real losses and overpressure.

The pressure path is compact, but its exact installed interfaces are unresolved

Pack section — buoy-mounted Hydro 1F proxy and direct closed-pack water-entry window



Technical evidence view generated from frozen scripted B-step source; not Case evidence. Unit is mm.

KNOWN

Leland 81121: 12 g CO₂, 3/8-24 puncture cartridge

KNOWN

Hydro 1F family uses V80040 water-sensitive bobbin

OPEN

Exact Hydro suffix and matching thread configuration

OPEN

Vendor-exact geometry, flow curve, patch interface, leak result

Topology: cartridge -> Hydro 1F family inflator -> custom 60 L buoy patch. No separate hose, tube, regulator, gauge, or pneumatic cylinder remains in this configuration.

The current configuration is custom-heavy and not fabrication-ready

79 MAKE

custom definitions requiring released manufacturing control

13 BUY

catalog or supplier-derived definitions

123

AP242/BREP/render state definitions delivered

Fabrication remains prohibited because the package does not contain approved dimensioned drawings, GD&T, tolerance stacks, process capability, torque/locking data, or validated tool access.

COTS-first objective

No complete, orderable commercial inflation module has passed geometry, interface, gas-capacity, and application gates.

NO PURCHASE OR FABRICATION AUTHORIZED

Release now depends on physical evidence and an architecture correction

1

Correct inflation architecture

Select a complete cartridge/inflator/buoy combination that passes capacity and rated-interface gates.

2

Define loads and release drawings

Control mission, impact, proof, life, tolerance, torque, material, finish, and inspection requirements.

3

Qualify the physical product

Wet opening, inflation, leak, proof/burst, buoyancy/stability, dynamic, fatigue, corrosion, and assembly trials.

Minimum physical test families

- inflation architecture and cold/hot delivery
- wet pack opening and extraction
- buoy proof, burst, leak, lift, and stability
- arm/lock/hardpoint/tether structural proof
- dynamic deployment and handling/drop
- fatigue, wear, corrosion, UV, humidity, storage
- realistic assembly, service, reset, and inspection

Digital pass; production release fails

DIGITAL PASS

- clean procedural build
- 180 occurrences per state
- hierarchy, occurrence identities, transforms, and valid solids preserved on reimport
- nominal envelope and mass properties
- endpoint exact-Boolean audit
- five-angle audit
- full 0-80 degree one-degree sweep

RELEASE FAIL

- 12 g / 60 L gas-capacity mismatch
- inflator suffix, exact geometry, ratings, and leak/flow unknown
- proxy and developmental softgood envelopes
- required load/strength/tolerance analyses incomplete
- manufacturing drawings and work instructions absent
- physical qualification absent
- AP242 bytes not reproducible across identical builds
- root assembly naming nonconformance remains open
- required screenshots and newer attachment placeholders absent

Do not release this configuration

Use the frozen digital package as a controlled engineering baseline. Correct the inflation architecture first; then define loads, complete manufacturing definition, qualify the physical product, and rerun independent review.

AUTHORIZATION: ENGINEERING REVIEW ONLY

No fabrication. No purchase. No qualification acceptance. No field use. No merge to main.

Next gate

Bring back one complete, orderable, source-supported inflation module and a controlled 60 L requirement. If none passes, stop with a measured architecture decision.

Digital baseline retained:

- STOWED / DEPLOYED AP242
- 92 unique parts
- 123 state definitions
- full 81-position audit